

VibeGuard: College FabLab 3D Printing Specification & Job Requisition

Project: VibeGuard — Edge AI Vibration Monitoring & Predictive Maintenance System

Institution: Jyothi Engineering College (Autonomous), Cheruthuruthy, Kerala

Department: Computer Science & Engineering / Electronics & Communication Engineering

Facility: College FabLab / 3D Printing & Prototyping Center

Lab Rate: ₹4.00 per gram (Standard College FabLab PLA/PETG Tariff)

Primary Contact: Amith Krishna Das (Hardware & Mechanical Lead)

1. Executive Summary for FabLab Operator / Technician

Note:

Context: This project investigates rotational unbalance and structural bearing vibration at low frequencies (10 Hz / 600 RPM). We need precision 3D-printed components to couple a miniature DC gear motor to an eccentric test mass and to rigidly mount an ADXL345 3-axis digital accelerometer.

We request the fabrication of **Two Essential Mechanical Parts** (and one optional safety shroud) on an FDM printer (Ender-3, Prusa MK3/MK4, or Bambu Lab):

Part ID	Component Name	Function	Material	In fill	Est. Mass	Est. Cost (@ ₹4/g)	Priority
FA-B-01	3 mm D-Shaft Eccentric	Slides onto N20 motor shaft; carries M3	PLA / PETG	80 %-10	3.5 g	₹14.00	CRITICAL (Mandatory)

Part ID	Component Name	Function	Material	In fill	Est. Mass	Est. Cost (@ ₹4/g)	Priority
	Rotor Arm	unbalance bolt & washers		0%			
FAB-02	ADXL345 Rigid Sensor Mounting Bracket	Rigidly couples accelerometer to baseplate without vibration damping	PLA	60%–80%	5.5 g	₹22.00	HIGH (Recommended)
FAB-03	Perforated Rotor Safety Shroud (Optional)	Protective containment cage over spinning 600 RPM rotor arm	PLA	20%	18.0 g	₹72.00	<i>Optional (Can use ₹0 clear cup)</i>
FAB-04	FAB-01 Spare / Backup Unit	Duplicate rotor arm for calibration bench backup	PLA / PETG	80%–100%	3.5 g	₹14.00	Recommended

- **Total Essential Print Package (FAB-01 × 2 + FAB-02): ~12.5 grams $\rightarrow$ Total Expected Cost: ₹50.00**
- **Total If Including Full Printed Shroud (FAB-01 × 2 + FAB-02 + FAB-03): ~30.5 grams $\rightarrow$ Total Expected Cost: ₹122.00**

2. Detailed Technical Specifications & CAD Dimensions

Part FAB-01: 3 mm D-Shaft Eccentric Rotor Arm (Cam)

Top View (mm)

```
+-----+
|  ( 0 )           ( 0 )  |
+-----+
|<---- 8 mm ---->|  |< 8 mm >|
    Shaft Hub       Mass Hub
|<----- 20.0 mm ----->|
      (Center-to-Center)
      Radius r = 12 mm
```

Exact Functional Dimensions

- **Motor Shaft Interface (D-Bore Hub):**

- * **Nominal Shaft Diameter:** 3.00 mm (N20 Micro Metal Gear Motor output shaft).
- * **D-Profile Geometry:** Circular bore of diameter 3.15 mm (accounting for FDM hole shrinkage) with a **flat chord face at 2.65 mm across the flat.**
- * **Hub Outer Diameter (\$OD\$):** 8.50 mm .
- * **Hub Height / Depth:** 9.00 mm (covers the full 9.5 mm flat of the N20 shaft).
- * **Set Screw Hole (Lateral):** 2.50 mm through-hole perpendicular to the flat face (to optionally tap an M3 grub screw for positive retention).

- **Connecting Arm Linkage:**

- * **Length (Shaft center to weight hole center):** 12.00 mm (governing dynamic eccentric radius r).
- * **Width:** 7.50 mm .
- * **Thickness:** 4.00 mm .

- **Eccentric Mass Mounting Hole:**

- * **Through-hole Diameter:** 3.20 mm (clearance for standard M3 bolt).
- * **Hub Outer Diameter around hole:** 8.00 mm .
- * **Counterbore / Nut Recess (Top Face):** Hexagonal recess of 5.60 mm across flats, depth 2.50 mm (to capture an M3 hex nut or Nyloc nut firmly without spinning).

Recommended Slicer Settings for FAB-01

- **Layer Height:** 0.16 mm or 0.20 mm.
- **Perimeters / Wall Loops:** 4 walls (minimum) — crucial so the shaft bore is solid plastic, not infill.
- **Top & Bottom Solid Layers:** 4 layers.
- **Infill:** 80% to 100% (Gyroid or Rectilinear). High density is required to withstand 10 Hz rotational centrifugal force without dynamic flexing.
- **Material:** PLA or PETG (Black, Grey, or Blue).

Part FAB-02: ADXL345 Rigid Accelerometer Mounting Bracket

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Front View (L-Bracket)
  | |  <-- 3.0 mm thick rigid upright
  | |  <-- 2x M3 holes spaced 15.0 mm apart (for ADXL345 board)
  | |
  ____| |____
  |_____|  <-- Stiff base flange (4.0 mm thick) with 2x wood screw
holes

```

Exact Functional Dimensions

- **Sensor Flange (Vertical Face):**
 - * **Height:** 22.00 mm , **Width:** 24.00 mm , **Thickness:** 3.50 mm .
 - * **Sensor Mounting Holes:** Two 3.20 mm diameter holes, spaced exactly 15.00 mm center-to-center horizontally, elevated 10.00 mm from the base.

- **Baseplate Mounting Flange (Horizontal Face):**

* **Depth:** 18.00 mm , **Width:** 24.00 mm , **Thickness:** 4.00 mm .

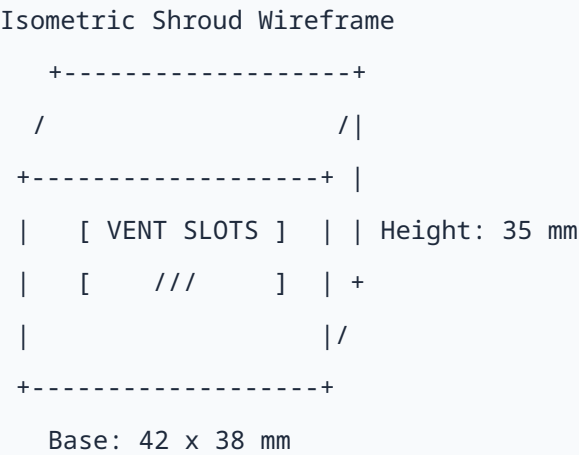
* **Base Screw Holes:** Two 3.50 mm countersunk holes spaced 16.00 mm apart for securing into the wooden baseplate with wood screws.

- **Stiffening Gussets:** Two triangular 45° gussets (3.0 mm thick) joining vertical and horizontal walls to prevent high-frequency cantilever resonance.

Recommended Slicer Settings for FAB-02

- **Layer Height:** 0.20 mm.
- **Infill:** 60% Gyroid.
- **Perimeters:** 3 to 4 walls.
- **Orientation:** Print resting on the flat base flange (no support required).

Part FAB-03: Perforated Safety Containment Shroud (Optional)



Exact Functional Dimensions

- **External Footprint:** $42.00\text{ mm} \times 38.00\text{ mm} \times 35.00\text{ mm}$ height.
- **Internal Clearance:** Guarantees $\geq 6.00\text{ mm}$ clearance around the 12 mm rotating arm + M3 bolt head at all 360° positions.

- **Wall Thickness:** 1.60 mm (4 perimeters with 0.4 mm nozzle).
 - **Observation & Airflow Slots:** Three $3\text{ mm} \times 20\text{ mm}$ vertical viewing slots on top and side to observe rotation without opening the guard.
 - **Mounting Lugs:** Two 10 mm exterior ears at base with 3.5 mm holes to screw directly onto the wood baseplate.
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3. Step-by-Step Instructions for FabLab Technician

- **CAD File Generation / Import:**

* Import the STL/STEP files into PrusaSlicer / Bambu Studio / Cura.

* If generating CAD from scratch in Fusion 360 / SolidWorks / Onshape, use the parametric sketches defined in Section 2.

- **Material Selection:**

* **Preferred:** Standard PLA (1.75 mm) — high Young's modulus ($E \approx 3.5\text{ GPa}$) ensures exceptional rigidity for vibration transfer.

* **Alternative:** PETG — good layer adhesion and impact resistance.

- **Print Execution:**

* Clean print bed with Isopropyl Alcohol (IPA) to ensure flat first layer without warping.

* Total print duration:

* FAB-01 (Rotor Arm): ~12 minutes per piece.

* FAB-02 (Sensor Bracket): ~18 minutes.

* Total runtime: **~40 minutes** for the essential set.

- **Post-Processing & Quality Check:**

* **Bore Fit Check:** Take a 3.0 mm steel rod or N20 motor shaft and test the D-profile fit. The rotor arm should slide on snugly by hand with zero radial wobble.

* **Deburring:** Clean any brim or elephants-foot brim from the bottom of the D-bore using a 3 mm hobby drill bit or craft blade.

4. Billing & Requisition Approval Slip

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JYOTHI ENGINEERING COLLEGE - FABLAB REQUISITION

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Project Title : VibeGuard (Edge AI Vibration Monitoring Rig)

Academic Course : PBCST504 / Final Year Embedded Project

Student Name : Amith Krishna Das & Team

Department : CSE / ECE

Items Ordered:

1. 3mm D-Shaft Eccentric Rotor Arm (FAB-01 x 2) : ~7.0 grams
2. ADXL345 Rigid Sensor Bracket (FAB-02 x 1) : ~5.5 grams
3. Optional Safety Shroud (FAB-03 x 1) : [] Check if requested (~18g)

Calculated Weight (Essential Set) : 12.5 grams

FabLab Tariff Rate : ₹4.00 per gram

ESTIMATED LAB FEE PAYABLE : ₹50.00 (or ₹122.00 w/ shroud)

Technician Signature : _____ Date: _____

Lab Supervisor Stamp : _____

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